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Long-term results after proton irradiation of uveal melanomas

Melanoma of the uveal tract is the only life-threatening tumor in the adult. Over the past three decades, definitive radiotherapy, which spares the eye and preserves vision, has gradually replaced enucleation as the preferred treatment for these tumors. Two major techniques are available. One involves surgical placement of radioactive plaques on the sclera over the area of the tumor (brachytherapy) [1–3]. The second involves external beam irradiation with charged particles such as protons [4] and helium ions [5]. The latter approach, introduced in 1975 as a collaborative effort of the Massachusetts Eye and Ear Infirmary (MEEI), the Massachusetts General Hospital, and the Harvard Cyclotron Laboratory [6], was expected to improve local control and reduce the complications of brachytherapy. Theoretical advantages of charged particle irradiation are based on the physical characteristics of protons and helium ions, which make possible uniform and highly localized dose distributions [7]. In particular, eyes with tumors close to critical structures such as the macula and optic nerve could be irradiated and still maintain visual potential, because the dose is sharply reduced outside the target volume [8]. Today, 12 proton beam facilities are used for the treatment of uveal melanomas in North America, Europe, South Africa, and Japan, and over 6000 patients have been treated worldwide. By the end of 1995, 2242 patients had been treated at Harvard, and in this paper we present long-term follow-up statistics for the earliest patients, all of whom were treated more than a decade ago. Although visual acuity continues to decline with extended follow-up, enucleations are uncommon, local control is well maintained, and annual metastasis rates remain low throughout the decade following irradiation.

Treatment protocol. Patients were treated with a total dose of 70 Co Gy equivalent (CGE) delivered in five equal treatments over 7–10 days. Surgical placement of

tantalum rings aided localization of the tumor during irradiation. During surgery, edges of the tumor were marked with blue pencil, and four tantalum rings 2.5 mm in diameter were sutured to the sclera to outline the tumor. No operation was necessary for tumors confined to the peripheral choroid, ciliary body and iris; in such cases the margins of the lesion in relation to anatomic landmarks of the iris and conjunctiva were defined by transillumination on the ocular surface.

Patient characteristics. A total of 487 patients (from the USA) underwent proton irradiation at the Harvard cyclotron between July 1975 and June 1984. Patients ranged in age from 14 to 85 years (median 58 years). The median largest tumor diameter was 13.5 mm (maximum 28.0 mm), and median tumor height as measured by ultrasound was 5.5 mm (maximum 15.0 mm). Thirty percent of patients had large melanomas (defined as greater than 15.0 mm in diameter and 5.0 mm in height). Eight patients had bilateral involvement. Extrascleral extension was observed in 13 patients (3%). In approximately a quarter (24%) of patients the anterior margin extended into the ciliary body; tumors involving only the ciliary body were relatively uncommon (2%). Approximately two-thirds of the patients had tumors extending posteriorly to within 3 mm of the optic disc, macula or both structures.

Clinical follow-up. Patients were examined on a regular basis at the MEEI or by contact with the referring ophthalmologist up to October 1995. Two hundred and ninety-five of the original 487 patients were alive and free of metastasis as of the most recent contact; 25% had died from metastasis and 15% from other causes. The median duration of follow-up among survivors was 12.2 years; the majority of these patients (86%) had follow-up in 1994/1995.

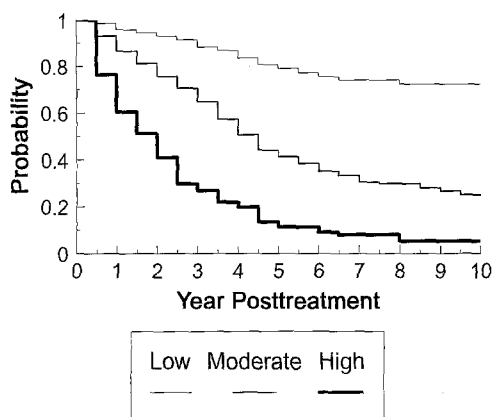


Fig. 1 Probability of retaining visual acuity of 20/200 or better by number of years since treatment

Local tumor control. Local tumor control was maintained in 97% of patients through 10 years of follow-up. Of the 12 documented or suspected cases of tumor regrowth, the majority (8) represented marginal recurrences. Patients with recurrence had over three times the risk of metastasis compared to other patients, controlling for other prognostic indicators (see below).

Eye retention. The overall probability of retaining the eye after proton beam irradiation was 90% at 10 years postirradiation. Three percent of eyes were enucleated during posttreatment year 1, and the annual rate dropped to 1% by the 4 year. Eyes with large tumors were at highest risk: the 10-year rate of enucleation was 7% for tumors under 13 mm in diameter, compared to 20% when the tumor was over 17 mm (approximately 15% of patients). The leading contributor to enucleation of the eye, accounting for over half of all treatment failures, was neovascular glaucoma.

Visual outcome. The most recent visual acuity measurement was obtained an average of 8 years posttreatment. Forty percent of patients had an acuity of 20/200 or better, and 18% an acuity of 20/50 or better, on the most recent examination. Leading factors associated with a less favorable visual outcome were proximity of the tumor to the optic nerve or fovea and increasing tumor elevation. The visual prognosis was also worse for patients with poorer pretreatment vision or macular detachment at presentation. Stratifying patients by risk group (based on categories of tumor height and posterior location), 79% and 72% of "low-risk" patients retained useful (20/200 or better) vision at 5 and 10 years, respectively, postirradiation (Fig. 1). However, few "high-risk" patients (5%) retained functional vision 10 years after treatment.

Complications. Leading complications that threaten vision after intraocular irradiation include radiation

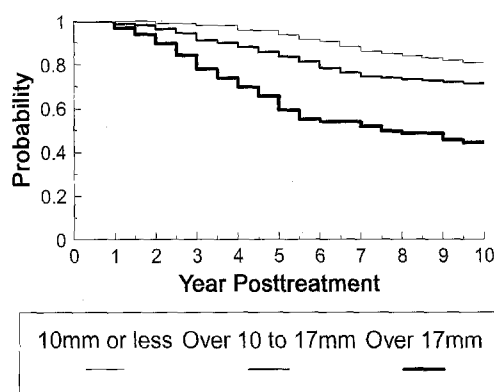


Fig. 2 Melanoma-related survival probability by largest tumor diameter

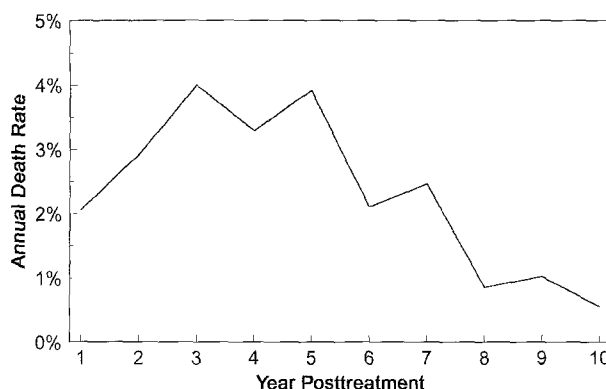


Fig. 3 Annual rate of death from metastatic melanoma by number of years since treatment

retinopathy and papillopathy, neovascular glaucoma, and cataract. *Radiation vasculopathy* occurred in 34% of treated eyes during the 10-year period of observation and was seldom observed outside of the irradiated area. The overall incidence of radiation papillopathy was low (14% by 10 years), except when the tumor abutted the optic disc (in 7% of patients: 66% by 10 years). Eyes with large anterior lesions were at risk of cataract and neovascular glaucoma. *Neovascular glaucoma* developed in 15% of patients by 10 years postirradiation. Eyes with highly elevated tumors, or receiving substantial radiation doses to the anterior segment, were at greatest risk. Incidence of *radiation cataract* increased linearly with radiation dose to the anterior segment: the 3-year rate of any opacification was 79% in those with high exposure (20% of patients), compared to 27% when the dose was minimal (50% of patients). In selected cases, cataract extraction with a posterior chamber intraocular lens improved the visual status.

Disease-free survival. Disease-free survival rates were 83% and 76% at 5 and 10 years, respectively, postirradiation.

tion (Fig. 2). Annual death rates from metastasis peaked in the 3rd and the 5th year postirradiation (4.0% and 3.9%, respectively) and declined thereafter to 0.6% by year 10 (Fig. 3). The longest interval before metastasis in this series was 13 years. The primary site of metastasis was the liver. By far the leading independent predictor of melanoma-related death was tumor diameter: death rates in patients with tumors over 17 mm were over seven times higher than in patients with small tumors (10 mm or smaller). Other characteristics associated with death from melanoma were extrascleral tumor extension, tumor extension anterior to the equator, greater patient age, and tumor involvement of the optic disc.

Summary. Over two decades of experience in Boston and at other centers worldwide confirm the efficacy of charged particle irradiation for treating tumors in the eye. Among 487 of the earliest patients treated by this modality, local control was achieved in 97%, and many retained longstanding useful vision in the 10-year period following treatment. Although complication rates were inevitably high in certain patient subgroups, a large majority retained the eye after therapy. Approximately

three-quarters of patients survived without metastasis for at least 10 years postirradiation. These survival data are comparable to previously published studies of patients undergoing enucleation, supporting results of nonrandomized trials which show no survival advantage associated with removal of the eye [9].

Future directions. In a NIH-sponsored randomized clinical trial, we are exploring the safety and efficacy of lowering radiation dosages to minimize complication rates and improve visual results. In addition, we have recently initiated a randomized clinical trial to determine whether interferon alpha-2a, when administered adjunctively, may prolong disease-free survival. Phase I studies are planned to evaluate the role of hyperbaric oxygen in the treatment of radiation papillopathy and of photodynamic therapy as an adjunct to protons in the treatment of selected tumors. Studies in molecular genetics are under way to identify the antecedents of intraocular melanoma at the genetic level. A planned epidemiologic case-control study should contribute further insight into environmental and thus preventable risk factors for these tumors.

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